

TIC 349701853

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 30.342709 d
Epoch (t0) 2395.4286 BTJD
Duration 0.1200 d
Depth 28.445 % (284,452 ppm)
SNR 343.5

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.03 (gate ≥ 0.50)
Robust depth sigma 252.5 (gate ≥ 7.0)
Odd/even z-score 0.87 (consistent)
Secondary eclipse detected, phase 0.33, 10.0 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 80 light-curve dip in this sector: 149.7 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 51.05 (gate ≥ 5.0) Compactness 0.62 (gate ≥ 0.50)
Dimming centroid offset 0.31 px (6 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 349701859
1.07 px away, dTmag -0.58 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

